Course Six

Google Advanced Data Analytics Capstone

Instructions

Use this PACE strategy document to record your decisions and reflections as a data professional as you work through the capstone project. As a reminder, this document is a resource guide that you can reference in the future and a space to help guide your responses and reflections posed at various points throughout the project.

Portfolio Project Recap

Many of the goals you accomplished in your individual course portfolio projects are incorporated into the Advanced Data Analytics capstone project including:

- Create a project proposal
- Demonstrate understanding of the form and function of Python
- Show how data professionals leverage Python to load, explore, extract, and organize information through custom functions
- Demonstrate understanding of how to organize and analyze a dataset to find the “story”
- Create a Jupyter notebook for exploratory data analysis (EDA)
- Create visualization(s) using Tableau
- Use Python to compute descriptive statistics and conduct a hypothesis test
- Build a multiple linear regression model with ANOVA testing
- Evaluate the model
- Demonstrate the ability to use a notebook environment to create a series of machine learning models on a dataset to solve a problem
- Articulate findings in an executive summary for external stakeholders

Project proposal

Salifort Motors Employee Retention

Overview

The HR team at Salifort Motors are developing an initiative to increase employee retention and satisfaction levels. They need a data analyst to examine the data they have collected from employee surveys and create a model capable of accurately predicting whether an employee will leave the company or not based on a variety of factors.

Milestones	Tasks	PACE stages
1	**Establish project workflow and create a project proposal.**	*Plan*
2	**Explore the available data for familiarity and compile summary information about it.**	*Plan, Analyze*
3	**Clean and prepare the data for analysis.**	*Plan*
4	**Complete an exploratory analysis of the available data.**	*Analyze*
5	**Construct visualizations to accompany the exploratory analysis to assist the HR team's understanding of the variables and distribution of data.**	*Analyze, Construct*
6	**Build a logistic regression model.**	*Analyze, Construct*
7	**Build one or more machine learning models.**	*Analyze, Construct*
8	**Assess all models using validation data and select the champion model.**	*Execute*
9	**Communicate final insights with the HR team using an executive summary.**	*Execute*

Data Project Questions & Considerations

PACE: Plan Stage

Foundations of Data Science

- Who is your audience for this project?

  The audience for this project is the HR team at Salifort Motors and any other relevant stakeholders for Salifort Motors.

- What are you trying to solve or accomplish? And, what do you anticipate the impact of this work will be on the larger business need?

  I am trying to solve the business' problem of reducing the rate of employees leaving the company and improving employee satisfaction. The HR team needs a model that is capable of accurately predicting whether an employee will stay with the company or leave based on a variety of variables collected using an employee survey. This project will help the business retain employees by identifying what factors make employees more at risk to leave the company. This will allow the company to implement interventions to improve employee satisfaction. Additionally, it will lead to cost savings as the company invests heavily in training new employees.

- What questions need to be asked or answered?
  - *What rate of error is acceptable for this company with the model?*
  - *What variables are included in the employee survey and do these encompass a complete picture of why or why not an employee would stay with the company?*
  - *Is the data complete and clean?*
  - *Is it possible to obtain any additional data if it's required?*
  - *What evaluation criteria will I use to determine whether a model is successful?*

- What resources are required to complete this project?

  I will utilize previous examples of work in my portfolio, previous course labs, and Google to look up accurate syntax and functions.

- What are the deliverables that will need to be created over the course of this project?

  I need to create a project proposal, an exploratory data analysis with data visualizations, a complete and accurate python lab with at least one model to predict employee retention, and an executive summary.

- How can you best prepare to understand and organize the provided information?
  I can best prepare to understand and organize the provided information by rereading all the given instructions to ensure I know what is being asked of me and reflecting on the flow of the project using the project proposal. When I don't understand, I can research solutions to improve my understanding.
- What follow-along and self-review codebooks will help you perform this work?
  I have found using Google to describe what I want to do to be helpful to find relevant and accurate links to describe how to perform the needed work. I use the Python Graph Gallery to help with choosing which visualizations fit my needs. Working alongside my previous lab work for this course using the new context has also been helpful.
  https://python-graph-gallery.com/
- What are a couple additional activities a resourceful learner would perform before starting to code?
  It would be useful to review this entire strategy document, particularly the planning section. Additionally, reviewing what libraries you need to import to the lab for the jobs you want to accomplish would be a good starting point before beginning to code.

Go Beyond the Numbers: Translate Data into Insights

- What are the data columns and variables and which ones are most relevant to your deliverable?
  The variable "left" will be my dependent variable (y) for the model. The other variables are my independent variables (x) and will contribute to the predictive value of my model. Some of these may be more useful than others, particularly "satisfaction_level".

- What units are your variables in?
  - *Satisfaction_level: a decimal number score to the hundredth place between 0 and 1*
  - *Last_evaluation: a decimal number score to the hundredth place between 0 and 1*
  - *Number_project: a whole number with 0 as the lowest possible value to describe how many projects an employee has contributed to*
  - *Average_monthly_hours: the average number of hours an employee worked per month*
  - *Time_spend_company: measured in number of years the employee has worked for the company*
  - *Work_accident: binary value with 0 as no and 1 as yes*
  - *Left: binary value with 0 as no (employee retained) and 1 as yes (employee left)*
  - *Promotion_last_5years: binary value with 0 as no and 1 as yes*

- *Department: name of the department (string)*
- *Salary: category descriptor of low, medium, or high (string)*

- What are your initial presumptions about the data that can inform your EDA, knowing you will need to confirm or deny with your future findings?
- *I initially presume that employees that are happier or highly paid are less likely to leave the company. Right away I can see there is plenty of opportunity to create some engineered features. This proved helpful in my previous logistic modeling project. This is something I can do during the EDA stage and then, also preparing the entirety of data for a logistic regression by checking assumptions and making a copy of the dataset without outliers. These engineered features could possibly be part of the dataframe for that and the other one for machine learning models.*
- Is there any missing or incomplete data?
- *The dataset has no missing values. It seems the biggest issue is the duplicate rows and messy column names.*
- Are all pieces of this dataset in the same format?
  No, most of the variables are integers (int64). Two of the variables (department and salary) are strings.
- Which EDA practices will be required to begin this project?
  The dataset needs to be thoroughly cleaned and checked for missing and inaccurate values. Engineered features may be useful for this project. I will need to determine how to resolve missing values before creating models.

The Power of Statistics

- What is the main purpose of this project?
  The main purpose of this project is to create a model that is able to predict whether an employee will leave the company or not.
- What is your research question for this project?
  Which employees will stay or leave a company based on a variety of variables?
  How can we increase employee retention and satisfaction?
- What is the importance of random sampling? In this case, what is an example of sampling bias that might occur if you didn't use random sampling?

Random sampling is important because it means the data represents the larger population of all employees at the company . This allows the company to apply the model to future groups of employees and identify strategies to improve retention for all employees at their company. An example of sampling bias without random sampling would be to either survey only the employees who are leaving the company or only those who are staying with the company. This would not provide as reliable insight to the company as surveying a random group pulled from all employees.

Regression Analysis: Simplify Complex Data Relationships

- Who are your stakeholders for this project?
  The stakeholders for this project are the Salifort Motors' HR team and any other relevant employees or stakeholders for the company.
- What are you trying to solve or accomplish?
  I'm trying to make a successful model at predicting whether an employee will leave the company or not.
- What are your initial observations when you explore the data?
  My first observation is that it includes many factors but the data does not completely encompass possible reasons why an employee has left the company. In the future, it may be useful to pursue data from exit surveys to investigate why employees have left the company.
- What resources do you find yourself using as you complete this stage?
  I have not used any resources yet, but plan to use my previous python labs from my Waze portfolio project, in which I followed a similar plan to the project proposal I set up for this project.
- Do you have any ethical considerations at this stage?

  I don't have any ethical concerns at this time because the data is anonymized, so all identifying personal information has been removed. Additionally, this project is aimed at improving employees' experiences and satisfaction levels, not for hiring purposes or for making decisions about employment status.

The Nuts and Bolts of Machine Learning

- What am I trying to solve?
  I am trying to create a model that will predict whether an employee will leave the company, so the company can increase employee retention and satisfaction.

- What resources do you find yourself using as you complete this stage?

  I have not used resources yet, but my previous Waze project will be invaluable for assisting with specific Python syntax and guiding me through the process.
- Is my data reliable?

  The data comes directly from the company. I can assume it is as reliable for this project as it can be.
- Do you have any additional ethical considerations in this stage?

  I'm not concerned about anything at this point, the variables included about employees are limited and don't include any that are connected to protected classes of individuals, so the model could not potentially be used to discriminate against these individuals. In addition, the variables are things that the company would not have access to until they have already been hired, so it could not be used to potentially limit individuals during the hiring process. The company plans to use the model to improve employee retention by improving their satisfaction. This will improve life for the employees and outcomes for the company. It does not lead to harm for employees or the company.
- What data do I need/would I like to see in a perfect world to answer this question?

  In a perfect world, I would have access to every possible variable for why an employee may leave a company. This, however, could lead to the model being used for questionable purposes and cause ethical concerns, and it is unrealistic because even with insights from exit interviews, people will lie or omit certain details about their experience.
- What data do I have/can I get?

  I have limited data for this project. I have data from an employee survey answered by 15,000 employees, which is a substantial amount of data values and should allow for splitting the data into train, validation and test sets. The dataset has 10 variables, including one variable that includes whether the employee has left or is retained within the company. I believe it will be enough to lead to valuable insights.
- What metric should I use to evaluate success of my business objective? Why?

  I will use a combination of F1, accuracy, precision, and recall scores to evaluate the models' success and choose the champion model to present to the company. These will identify what the company can predict related to future performance of the model and will reveal the rates of false negatives and false positives created by the model.

Data Project Questions & Considerations

PACE: Analyze Stage

Foundations of Data Science

- Will the available information be sufficient to achieve the goal based on your intuition and the analysis of the variables?

  I believe the available data will provide valuable insights to the company and likely will be able to create a model that is informative for the company's purposes.

Go Beyond the Numbers: Translate Data into Insights

- What steps need to be taken to perform EDA in the most effective way to achieve the project goal?

  Variables need to be explored and prepared for analysis. I need to identify related variables to ensure I select the best possible combination for the logistic regression test. Finally, I need to examine how variables relate to "left" using a correlation heatmap.

- Do you need to add more data using the EDA practice of joining? What type of structuring needs to be done to this dataset, such as filtering, sorting, etc.?

  I do not need to use joining at this stage of the project. I used filtering by certain variables to examine relationships between another variable and "left".

- What initial assumptions do you have about the types of visualizations that might best be suited for the intended audience?

  Visualizations that suit this stage of the project include bar graphs, histograms, pie charts and scatter plots.

The Power of Statistics

- Why are descriptive statistics useful?

  Descriptive statistics were useful for the analysis because I used them to help identify outliers. It also helped me better understand the variables' spread and the average and ranges of the variables.

- What is the difference between the null hypothesis and the alternative hypothesis?

The null hypothesis is the default statement that states nothing has changed or is statistically significant. The alternative hypothesis is the proposed change or difference between two groups. It states that this difference is statistically significant.

Regression Analysis: Simplify Complex Data Relationships

- What are some purposes of EDA before constructing a multiple linear regression model?

  Completing an EDA helps aid with selection of variables to use in the model. It helps you understand how the variables are related and identify if variables are multicollinear. It is the stage where you initially check for some of the assumptions to create a useful model.

- Do you have any ethical considerations at this stage?

  I do not have any ethical considerations in this stage, seeing as the data is limited and the model will be used to improve individuals' work lives.

The Nuts and Bolts of Machine Learning

- What am I trying to solve? Does it still work? Does the plan need revising?
  I need to create either a logistic regression model or a machine learning model that can identify which employees are at risk of leaving the company based on a collection of variables that the company has control over. The company wants to use this model to best implement interventions to reduce employees leaving the company after receiving valuable and costly training. If employees are able to be better retained, the company will save money and retain talent. I expect the plan will still work and does not need revising at this time.
- Does the data break the assumptions of the model? Is that ok, or unacceptable?
  The data does break some of the assumptions of a logistic regression model but not the machine learning models. I prepared an additional dataset using a subset of data without outliers for use in the logistic regression. That has resolved the issue of extreme outliers. Some assumptions have been met including the outcome variable "left" is categorical, observations are independent (addressed by dropping duplicates), and sufficiently large sample size. Some of the variables are multicollinear; I will have to be cautious when selecting the X variables to use for the model. I am unable to test for a linear relationship between each X variable and the logit of the outcome variable until the model has been completed.

- Why did you select the X variables you did?

I am going to run multiple models and compare. This data is appropriate for a binary logistic regression model, so I can run a logistic regression model and tree-based machine learning models, including decision tree and random forest. For the logistic regression, I need to be cautious about multicollinearity among the X variables. When variables have a Pearson correlation coefficient value greater than the absolute value of 0.7 with each other, these variables are strongly multicollinear and I will select the variable that has the highest correlation with 'left', which is the binary target variable. I am using the df_no_outliers for the logistic regression model.

The following variables are strongly multicollinear:

- hours per month & hard worker (0.75)

- hours per month & overworked (0.83)

- tenure & total hours (0.76)

Out of these variables, I have selected hard worker, overworked and tenure to be used in the X variables. Hours per month and total hours must be dropped from df_no_outliers. The X variables will be hard worker, overworked, tenure, performance score, projects, work accident, recent promotion, salary, full time, high performer, and under appreciated. The y variable will be left.

- What are some purposes of EDA before constructing a model?

  Doing an EDA before constructing a model is important because it allows you to become familiar with the data and context, prepare the data adequately for the model and check for the assumptions that can be identified before building the model.

- What has the EDA told you?

  The EDA was helpful for insights about both the employees who left the company and those who were retained. There were clear patterns identifying low employee satisfaction, high hours per week, and higher numbers of projects as important demographics for employees more likely to leave the company.

- What resources do you find yourself using as you complete this stage?

  I have utilized previous labs to repeat procedures for an EDA and check correct syntax to create the visualizations I needed.

- Do you have any ethical considerations at this stage?

  I do not have any ethical considerations at this time.

Data Project Questions & Considerations

PACE: Construct Stage

Foundations of Data Science

- Do any data variables averages look unusual?

  None of the variables' averages look unusual. I already dealt with the outliers in the tenure category.
- How many vendors, organizations or groupings are included in this total data?

  That question was not answered by the company. I would have to inquire further to find out.

Go Beyond the Numbers: Translate Data into Insights

- What data visualizations, machine learning algorithms, or other data outputs will need to be built in order to complete the project goals?

  To complete the project goals, I need to make a binary logistic regression model. I can either make a logistic regression model or a tree-based machine learning model. Additionally, I will need the scores related to the chosen model's expected performance for the company, and data visualizations to explain how the model performs for the stakeholders and to show the most important features in the model. This information will need to be presented in a concise executive summary.
- What processes need to be performed in order to build the necessary data visualizations?

  To build satisfactory data visualizations, I can use seaborn and sklearn.metrics on python. I need to build the models first and choose the champion model based on their performance. Then, I can use the test dataset on the model to see how it is expected to perform for the company. I can create the visualizations using this information.
- Which variables are most applicable for the visualizations in this data project?

  The most important variables seem to be left, tenure, salary and the variable I created- under_appreciated. Satisfaction scores also were related to left and found to be a very important factor for many employees, but not all segments of the population.
- Going back to the Plan stage, how do you plan to deal with the missing data (if any)?

I don't believe the dataset had any missing values to deal with. It did have a small percentage of outliers and duplicates, which I decided to deal with by dropping from the dataset.

The Power of Statistics

- How did you formulate your null hypothesis and alternative hypothesis?
  I did not do a test like this when constructing the models, but while preparing the data for modelling. I decided it would be useful when determining if I could drop a variable- employee department. My null hypothesis was simply that the employee department had no impact on the rate of leaving the company, and the alternative hypothesis was the opposite: that the rate of leaving segmented by employee departments did differ and were statistically significant.
- What conclusion can be drawn from the hypothesis test?

  My p-value was very high, around 0.95, so I determined that I could not reject the null hypothesis. Department type did not have a statistically significant impact on the rate of employees leaving. I decided to drop the variable from the data frame because leaving it had minimal gains and the risk of overfitting for the model by creating many variables encoding department type into a numerical data type was much greater.

Regression Analysis: Simplify Complex Data Relationships

- Can you improve it? Is there anything you would change about the model?
  The initial logistic regression model was a good model. It did an acceptable job meeting the company's task but I was able to build more successful machine learning models. I could improve the logistic regression model by adjusting parameters and trying different combinations of variables, adding additional data, or creating more engineered variables. I am not moving forward with that though, because my initial attempts at the machine learning models were both more successful.

The Nuts and Bolts of Machine Learning

- Is there a problem? Can it be fixed? If so, how?

I had an issue with the parameter setup for the random forest model . The initial best score came out as 0. This was unacceptable to continue with a model with that score, since a random guess even would be more accurate. I was able to adjust the parameters to achieve a successful score of 0.9087, which showed the model would more likely be effective. I realized that I was only programming the model to train on a single selected parameter setting, rather than a range to choose from, and I had set a score to 1, so the model was not given any opportunity to repeat and learn from the data.

- Which independent variables did you choose for the model, and why?

  The independent variables were hard worker, overworked, tenure, performance score, projects, work accident, recent promotion, salary, full time, high performer, and under appreciated. I only limited the variable selection by not including variables that were highly multicollinear and by dropping employee department (not correlated and risk of overfitting to make numerical and causing 10 new variables to be created) and satisfaction score (risk of employer not having the data for all employees in the future and questions of accuracy and reliability of this subjective score).

- How well does your model fit the data? (What is my model's validation score?)

  All my models performed well with the data with both the random forest and XGboost being excellent models. Recall is where the greatest "risk" of not identifying individuals as leaving the company lies. I want to have fewer employees predicted to stay when they will actually leave, as reducing the number there should have the biggest impact on retention rates in the company. I want the model to do a good job at correctly predicting employees who will leave and then actually leave. This will help the company find potential subcategories of employees to target positive interventions towards. Precision is less important because in this case it's more risky to not capture employees who will leave than to let some that will stay incorrectly be predicted as employees who will leave. I need to be particularly careful of evaluating only accuracy because my model is based on a dataset where employees leaving are the minority and the majority stay. It could get a decent accuracy score, even if it only guessed all employees would stay, which would not help me meet the company's goals for the project. I believe the random forest model is my champion model here, even if it gets quite lower precision scores than the XGBoost model. It has the highest recall score and is the best model to identify employees who could potentially leave, but does not falsely identify too many employees as leaving when they will stay which could lead to unnecessary interventions and expenses, and still has an excellent accuracy score and good precision and F-1 scores.

Model	Accuracy	Precision	Recall	F1- Score
Logistic Regression	0.926829	0.749288	0.787425	0.767883
Random Forest	0.964554	0.868235	0.927136	0.896719
XGBoost	0.973728	0.944297	0.894472	0.918710

- Can you improve it? Is there anything you would change about the model?
  It could potentially be improved, but it would take more computational resources and time. This would be worthwhile if the company needed a higher performing model. In this case, over identification of employees who will leave has minimal risks. The company is going to implement this model to try to decrease the percentage of employees leaving their company so they can save money and valuable investments in employee training. The company will implement strategies to decrease employees leaving based on the variables shown to be of most importance by the model. These changes will likely benefit all employees, even those that the model deems will leave the company but actually would have stayed anyway even without the changes.
- Do you have any ethical considerations at this stage?
  I do not have any ethical considerations at this stage, given the few risks involved. When the model identifies a person as "left" and that is true, the employee may have interventions targeted towards them that will reduce the chance of them leaving, which may be beneficial to the employee and unlikely to cause them harm. When the model identifies a person as "left" and that is false, the employee was going to stay but is subject to interventions and improvements, which will likely improve their experience at the company and are unlikely to cause them harm. When the model identifies a person as "stayed" and that is true, the employee's work life will probably go unchanged, though some of the interventions and improvements may also benefit all employees at the company. When the model identifies a person as "stayed" and that is false, the employee is not targeted for potential interventions they could benefit from and will likely leave the company, but this is the same fate as not having a model, so I would anticipate

Data Project Questions & Considerations

PACE: Execute Stage

Foundations of Data Science

- Given your current knowledge of the data, what would you initially recommend to your manager to investigate further prior to performing an exploratory data analysis?

  I believe the current data is sufficient for the job at hand. It was able to create successful models. The EDA showed that a large portion of employees leaving were the most overworked and underappreciated within the business. The company should take steps to combat employee burnout and show appreciation to their employees.

- What data initially presents as containing anomalies?

  The only data I noticed that contained anomalies was the outliers within the tenure category.

- What additional types of data could strengthen this dataset?

  Other data that could strengthen this dataset includes more detailed performance ratings broken up into categories, satisfaction ratings broken down into smaller categories, date of last performance review, and amount of training the employees received if it varies by individual.

Go Beyond the Numbers: Translate Data into Insights

- What key insights emerged from your EDA and visualizations(s)?

  Sub-populations of employees at risk of leaving the company:

  1) *Overworked, unsatisfied employees that do not feel appreciated due to very high hours, high amounts of projects and lack of promotions within the last 5 years. These employees are unhappy with their work lives and are probably looking to leave the company for other employment where they will work less and have lower demands for the pay level.*

2) *Employees that are satisfied and perform well at their work but are burned out from very high hours, even up to twice the hours in a standard work week. These employees may leave the company in search of higher paying positions with lower hours or more benefits, possibly stepping up to higher status positions.*

3) *Employees that are unsatisfied and perform poorly at their jobs, perhaps the employees that left due to being fired from the company.*

4) *Employees that have 6 or more projects. These employees may have been looking for a job with less demands. Around half of all employees with 6 projects left, while all employees that had 7 projects left the company.*

5) *Employees that had only 2 projects. These employees might feel under utilized by the company or they are looking for more complex work. Around half of all employees with 2 projects ended up leaving the company.*

Sub-populations of employees who are more likely to be retained:

1) *Employees who have received promotions within the last 5 years.*

2) *Employees who are part-time, working less than 130 hours per month.*

3) *Employees with higher satisfaction scores that work a reasonable amount of hours per month.*

4) *Employees with between 3 and 5 projects.*

5) *Employees with the company for under 2 years or over 7 years.*

Employees across all departments are equally likely to leave the company. The difference in retention rates among departments is not statistically significant. Department is not a predictive variable to base a model on. The HR department can conclude that this evidence suggests that there is not a major issue among specific departments affecting employee retention rate. This also suggests the HR department can focus implementing solutions more broadly among the employee population than tailoring the approach to select departments.

- What business recommendations do you propose based on the visualization(s) built?

I would suggest employers cap the amount of projects employees can work on at one time to 5 so they aren't spread too thin and have to work extremely long hours to get their projects done. Between 3 and 5 projects at a time would be ideal for most employees. Additionally, the company should ensure employees receive sufficient time off for the hours worked to combat burn out. Data indicates employee burnout is pervasive throughout the company due to a culture of long hours and heavy workloads. The company should look into whether their benefit packages are sufficient and investigate whether employees have simple and confidential access and awareness of mental health care, such as online therapy. The company should research ways that they can reward employees for their work so they feel appreciated and satisfied. The company should consider surveying long-time employees who have worked over 7 years to find out factors that lead to loyalty to the company.

The Power of Statistics

- What key business insight(s) emerged from your A/B test?

Employees across all departments are equally likely to leave the company. The difference in retention rates among departments is not statistically significant. Department is not a predictive variable to base a model on.

- What business recommendations do you propose based on your results?

The HR department can conclude that this evidence suggests that there is not a major issue among specific departments affecting employee retention rate. This also suggests the HR department can focus implementing solutions more broadly among the employee population than tailoring the approach to select departments.

Regression Analysis: Simplify Complex Data Relationships

- To interpret model results, why is it important to interpret the beta coefficients?

  Examining the beta coefficients allows you to see the relationships the variables have with the target variable. Those variables that are closer to 0 have less correlation with the target variable. If the variable's coefficient is a negative number it has a negative correlation with the target variable, which becomes stronger as the number gets smaller. If the variable's coefficient is a positive number it has a positive correlation with the target variable, which becomes stronger as the number gets larger.

- What potential recommendations would you make to your manager/company?

  I would recommend a tree-based machine learning model be used instead of the logistic regression model. The machine learning models had higher validation scores and alterations to the regression model might lead to improvements in the scores but not likely such as big improvement as simply changing to the machine learning model. The machine learning model also takes less preparation of the data to successfully run, though it does take more computational resources.

- Do you think your model could be improved? Why or why not? How?

  The model potentially could have been improved by adjusting parameters, additional data, or creating additional engineered variables. Because the logistic regression model had significantly lower scores than the tree-based models, I decided not to move forward with that approach and used a tree-based model instead.

- Do you have any ethical considerations at this stage?

  Yes, I checked the potential impacts from all outcomes of the model (what happened if it predicted wrong) to consider if there were any ethical concerns of using the company using the model. I found that use of the model had minimal risks and potentially great benefits.

The Nuts and Bolts of Machine Learning

- What key insights emerged from your model(s)? What business recommendations do you propose based on the models built?

  The machine-learning model revealed significantly different insights than the exploratory data analysis. It would be beneficial for the company to keep both in mind when moving forward with implementing solutions to reducing employees leaving the company. The exploratory data analysis revealed the general profiles of employees most at risk of leaving the company and the sub-populations of employees who were likely to remain loyal to the company. Employees were more likely to leave the company when they had more than 6 projects and worked long monthly hours, particularly if they hadn't received a promotion within the last 5 years or weren't among the company's highest paid employees. This indicates the rate of employees leaving is being affected by burn out. This is a common issue across the workforce, so the HR department can turn towards other companies and examine what approaches have and have not been successful in the past to combat employee burnout. For example, the HR department could encourage a policy that employees have a max of 5 projects at a time. It also showed that employee departments have little to no correlation with employees leaving. This indicates the HR department can look to implement solutions more broadly rather than departmentally.

- What are the criteria for model selection?

  My criteria for selecting the model was based on validation data for all three of the models I created. I was looking for a model that performed well in regards to accuracy, precision, and the f1 score, but particularly recall. The company will use the model to identify groups of employees who are more at risk for leaving the company so they can prevent this. The recall score tells us how well the model performs at correctly identifying individuals who will leave the company. It is calculated by taking the number of correctly identified "left" employees divided by the total number of "left" employees. The recall score tells us that the model is able to correctly identify left employees 92.7% of the time. The confusion matrix tells us that only 29 "left" employees were missed by the model, which is a small number compared to the approximately 10,000 individuals in the dataframe.

- Does my model make sense? Are my final results acceptable?

  Obviously, the higher the recall score, the better the model will perform for the company's purposes. I do believe my model will do an adequate job for the company and putting it into use will help them determine interventions and reduce employee churn. It is able to capture most of the employees at

risk. With additional data variables, engineered features, or adapting the parameters, it is possible that the model could be refined to improve its performance if the company should require that.

- Were there any features that were not important at all? What if you take them out?

  There was a feature that was not important at all- type of department, and I took it out of the model because using it would have resulted in 10 extra variables for this data after changing it to numerical values and the risk of overfitting the model was greater than the benefits of including it in the model.

- What resources do you find yourself using as you complete this stage?

  I used previous projects that I have completed through this course to follow the process I used previously and correct python syntax.

- Is my model ethical?

  My model is ethical because use of the model will lead to minimal harm and potentially great benefit for the people whose data is being utilized and the company who will be using the model.

- When my model makes a mistake, what is happening? How does that translate to my use case?

  When the model identifies a person as "left" and that is true, the employee may have interventions targeted towards them that will reduce the chance of them leaving, which may be beneficial to the employee and unlikely to cause them harm. When the model identifies a person as "left" and that is false, the employee was going to stay but is subject to interventions and improvements, which will likely improve their experience at the company and are unlikely to cause them harm. When the model identifies a person as "stayed" and that is true, the employee's work life will probably go unchanged, though some of the interventions and improvements may also benefit all employees at the company. When the model identifies a person as "stayed" and that is false, the employee is not targeted for potential interventions they could benefit from and will likely leave the company, but this is the same fate as not having a model.